CSE 112 – Lab #10 Linked List using the STL list container – Linked Lists

Requirements: Using the STL list container, create the list on page 1157 (18 - 6), and **modify** the program as described below.

- Define a list object, and **populate** it using a loop as shown in 18-6, **but up to 200 by 20's**.
- Call a function to **display** the list and output the number of nodes in the list.
- Then, call a function to **add** to the list the numbers (10, 30, 50, 70, 90, 110, 130, 150, 170) inserting them in numerical order using a loop, an iterator, list.begin, and list.insert.
- Then call the **display** function to display the list again and the number of nodes.
- Then **reverse** the list and display it again, and output the number of nodes.
- Then call a function that **removes the ends** of the list, and then **redisplay** the list and the number of nodes.
- Use an **iterator** in the solution for adding the numbers and for output at a minimum.

All output is to be done in the output function. The output function that displays the list **must accept a string** for the message prior to displaying the list and the number of nodes.

Sample output function declaration: void display_list(list<int> &myList, **string** message);

Sample output function call: display_list(myList, "**The original list is**: ");

Output Format and contents:

```
"E:\RCBC\Spring 2022\CSE 112 100\Labs and Solutions CSE 112\L...

The original list is:
0 20 40 60 80 100 120 140 160 180
The number of list nodes is: 10


The list with the additional numbers is:
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180
The number of list nodes is: 19


The list in reverse is:
180 170 160 150 140 130 120 110 100 90 80 70 60 50 40 30 20 10 0
The number of list nodes is: 19


The list with the ends removed:
170 160 150 140 130 120 110 100 90 80 70 60 50 40 30 20 10
The number of list nodes is: 17


Process returned 0 (0x0)   execution time : 0.669 s
Press any key to continue.
```

Grading will be based on meeting all of the lab requirements, adherence to programming standards, operation and accurate output, and programming style including white space and indentation.

Assignments submitted after the due date will incur a grade penalty of three (3) points per day after the first day, and will not be accepted more than one (1) week late.